

<Mean-Variance Analysis – with no riskless asset>

$$\text{Min } \frac{1}{2} \sigma_p^2 = \frac{1}{2} \tilde{w}^T \Sigma \tilde{w} \quad (= \frac{1}{2} \sum_i \sum_j w_i w_j \sigma_{ij})$$

$$\text{Subject to } \tilde{w}^T \cdot \mu = \mu_p$$

$$\tilde{w}^T \cdot \mathbf{1} = 1$$

$$\text{where } \tilde{w} = \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix}, \quad \mu = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \end{pmatrix}, \quad \text{and } \mathbf{1} = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} \quad \text{are given information}$$

and $\mu_p \equiv$ target return (a scalar)

Assume a frictionless economy and unlimited short selling. Σ is nonsingular (invertible). $\tilde{w}^T \Sigma \tilde{w} > 0$ (positive definite) with $\tilde{w}^T \neq 0$ (at least one element $\neq 0$).

Forming the lagrangian, $\tilde{w}_p$ is the solution to the following

$$\text{Min } L = \frac{1}{2} \tilde{w}^T \Sigma \tilde{w} + \lambda (\mu_p - \tilde{w}^T \mu) + \gamma (1 - \tilde{w}^T \mathbf{1})$$

The First order condition (F.O.C)

$$\frac{\partial L}{\partial \tilde{w}} = \Sigma \tilde{w}_p - \lambda \mu - \gamma \mathbf{1} = 0 \quad (1)$$

$$\frac{\partial L}{\partial \lambda} = \mu_p - \tilde{w}_p^T \cdot \mu = 0 \quad (2)$$

$$\frac{\partial L}{\partial \gamma} = 1 - \tilde{w}_p^T \cdot \mathbf{1} = 0 \quad (3)$$

where 0 is a N vector of zeros in (1)

$$\text{From (1), } \underline{w}_p = \lambda \cdot \Sigma^{-1} \cdot \underline{\mu} + \gamma \cdot \Sigma^{-1} \cdot \mathbf{1} \quad (1)'$$

$$\text{From (2), } \mu_p = \underline{w}_p^T \cdot \underline{\mu} = \underline{\mu}^T \cdot \underline{w}_p = \lambda \left(\underline{\mu}^T \cdot \Sigma^{-1} \cdot \underline{\mu} \right) + \gamma \left(\underline{\mu}^T \cdot \Sigma^{-1} \cdot \mathbf{1} \right) \quad (2)'$$

$$\begin{array}{ccc} & B & A \\ \text{From (3), } 1 = \underline{w}_p^T \cdot \mathbf{1} = \mathbf{1}^T \cdot \underline{w}_p = \lambda \left(\mathbf{1}^T \cdot \Sigma^{-1} \cdot \underline{\mu} \right) + \gamma \left(\mathbf{1}^T \cdot \Sigma^{-1} \cdot \mathbf{1} \right) & & (3)' \\ & A & C \end{array}$$

Using (2)' and (3)',

$$\lambda = \frac{C \cdot \mu_p - A}{D}$$

$$\gamma = \frac{B - A \cdot \mu_p}{D}$$

$$A = \mathbf{1}^T \cdot \Sigma^{-1} \cdot \underline{\mu} = \underline{\mu}^T \cdot \Sigma^{-1} \cdot \mathbf{1}$$

$$\text{where } B = \underline{\mu}^T \cdot \Sigma^{-1} \cdot \underline{\mu} > 0$$

$$C = \mathbf{1}^T \cdot \Sigma^{-1} \cdot \mathbf{1} > 0$$

$$D = BC - A^2 > 0$$

- Note that the inverse of a positive definite matrix is positive definite (P.D.) so $B > 0$, $C > 0$.

We claim $D > 0$, since the inverse of a positive definite matrix is P.D.

$$\left(A\underline{\mu} - B\mathbf{1} \right)^T \Sigma^{-1} \left(A\underline{\mu} - B\mathbf{1} \right) \quad (4)$$

$$= A^2 \cdot B - B \cdot A \cdot A - A \cdot B \cdot A + B \cdot B \cdot C$$

$$= B \left(BC - A^2 \right) = B \cdot D$$

L.H.S of (4) is strictly positive since Σ^{-1} is positive definite $\Rightarrow D > 0$

Substituting λ and γ into Eq(1)', optimal weight of frontier portfolio

$$\tilde{w}_p = \left(\frac{C\mu_p - A}{D} \right) \Sigma^{-1} \cdot \tilde{\mu} + \left(\frac{B - A\mu_p}{D} \right) \Sigma^{-1} \cdot \mathbf{1} = g + h\mu_p$$

$$\text{where } g = \frac{1}{D} (B \cdot \Sigma^{-1} \cdot \mathbf{1} - A \cdot \Sigma^{-1} \cdot \tilde{\mu})$$

$$h = \frac{1}{D} (C \cdot \Sigma^{-1} \cdot \tilde{\mu} - A \cdot \Sigma^{-1} \cdot \mathbf{1})$$

Note that g and h are fixed given $\tilde{\mu}$ and Σ

***Proposition**

An arbitrary frontier portfolio, q can be generated by any two frontier portfolio, p_1 and p_2 . In other words, $\tilde{w}_q$ can be made by a linear combination of $\tilde{w}_{p_1}$ and $\tilde{w}_{p_2}$.

Proof :

Since $\mu_{p_1} \neq \mu_{p_2}$, there exists a constant, α such that

$$\mu_q = \alpha \cdot \mu_{p_1} + (1 - \alpha) \mu_{p_2}$$

Now

$$\begin{aligned} & \alpha \tilde{w}_{p_1} + (1 - \alpha) \tilde{w}_{p_2} \\ &= \alpha (g + h\mu_{p_1}) + (1 - \alpha) (g + h\mu_{p_2}) \\ &= g + \alpha h\mu_{p_1} + (1 - \alpha) h \cdot \mu_{p_2} \\ &= g + h [\alpha \mu_{p_1} + (1 - \alpha) \mu_{p_2}] \\ &= g + h\mu_q \\ &= \tilde{w}_q \end{aligned}$$

$\Rightarrow q$ can be represented by a linear combination of p_1 and p_2

***Relationship between μ_p and σ_p^2**

$$\sigma_p^2 = \tilde{w}_p^T \Sigma \tilde{w}_p = \tilde{w}_p^T \Sigma \left(\lambda \Sigma^{-1} \tilde{\mu} + \gamma \Sigma^{-1} \mathbf{1} \right) \quad (\text{From (1)'})$$

$$= \lambda \tilde{w}_p^T \Sigma \Sigma^{-1} \tilde{\mu} + \gamma \tilde{w}_p^T \Sigma \Sigma^{-1} \mathbf{1}$$

$$= \lambda \tilde{w}_p^T \tilde{\mu} + \gamma \tilde{w}_p^T \mathbf{1} = \lambda \mu_p + \gamma \cdot 1$$

$$= \left(\frac{C \cdot \mu_p - A}{D} \right) \mu_p + \left(\frac{B - A \mu_p}{D} \right)$$

$$= \frac{1}{D} \left[C \mu_p^2 - 2A \mu_p + B \right]$$

$$= \frac{1}{D} \left[C \left(\mu_p^2 - \frac{2A}{C} \mu_p + \frac{A^2}{C^2} \right) - \frac{A^2}{C} + B \right]$$

$$= \frac{1}{D} \left[C \left(\mu_p - \frac{A}{C} \right)^2 + \frac{D}{C} \right] \quad (\text{note: } D = BC - A^2)$$

$$\Rightarrow \sigma_p^2 = \frac{C}{D} \left(\mu_p - \frac{A}{C} \right)^2 + \frac{1}{C} \quad (5)$$

This is an equation of a parabola in $\mu - \sigma^2$ space; A hyperbola in $\mu - \sigma$ space

$\Rightarrow$ The variance of the MVP is $\frac{1}{C}$, and the expected return on the MVP is $\frac{A}{C}$

***Optimal weight of the MVP**

On the MVP, $\mu_p = \frac{A}{C}$

$$\text{so, } \lambda = \frac{C \cdot \mu_p - A}{D} = \frac{C \cdot \frac{A}{C} - A}{D} = 0$$

$$\gamma = \frac{B - A \cdot \mu_p}{D} = \frac{B - A \cdot \frac{A}{C}}{D} = \frac{1}{C} \quad (\text{note } D = BC - A^2)$$

Now, the optimal weight can be represented as (from(1)')

$$\tilde{w}_{MVP} = \lambda \Sigma^{-1} \tilde{\mu} + \gamma \Sigma^{-1} \mathbf{1}$$

$$= \frac{1}{C} \Sigma^{-1} \cdot \mathbf{1} \text{ or } \frac{\Sigma^{-1} \mathbf{1}}{\mathbf{1}^T \Sigma^{-1} \mathbf{1}}$$

***Covariance between any portfolio , P (not necessarily on the frontier)
and the MVP**

$$\begin{aligned}
 \sigma_{p,MVP} &= \mathbf{w}_{MVP}^T \cdot \Sigma \mathbf{w}_p \\
 &= \left(\frac{1}{C} \Sigma^{-1} \mathbf{1} \right)^T \Sigma \mathbf{w}_p \\
 &= \frac{1}{C} \mathbf{1}^T \Sigma^{-1} \cdot \Sigma \mathbf{w}_p = \frac{1}{C} \mathbf{1}^T \mathbf{w}_p = \frac{1}{C} = \sigma_{MVP}^2
 \end{aligned}$$

**<Mean-Variance Portfolio analysis –
with the riskless asset>**

$P \equiv$ frontier portfolio containing $(N+1)$ assets

$$\underset{\{\tilde{w}, \lambda\}}{\text{Min}} \quad L = \frac{1}{2} \tilde{w}^T \Sigma \tilde{w}$$

$$\text{s.t.} \quad \tilde{w}^T \tilde{\mu} + (1 - \tilde{w}^T \mathbf{1}) R_f = \mu_p$$

Note $\tilde{w}$ (n -vector): weights of risky assets

$\tilde{w}^T \mathbf{1}$: total weight of risky assets

$1 - \tilde{w}^T \mathbf{1}$: weight of the riskless assets

Forming the lagrangian,

$$\underset{\{\tilde{w}, \lambda\}}{\text{Min}} \quad L = \frac{1}{2} \tilde{w}^T \Sigma \tilde{w} + \lambda \left[\mu_p - \tilde{w}^T \tilde{\mu} - (1 - \tilde{w}^T \mathbf{1}) R_f \right]$$

$$\frac{\partial L}{\partial \tilde{w}} = \Sigma \tilde{w}_p - \lambda (\tilde{\mu} - \mathbf{1} R_f) = 0 \quad (6)$$

$$\frac{\partial L}{\partial \lambda} = \mu_p - \tilde{w}_p^T \tilde{\mu} - (1 - \tilde{w}_p^T \mathbf{1}) R_f = 0 \quad (7)$$

From (6),

$$\tilde{w}_p = \lambda \Sigma^{-1} (\tilde{\mu} - \mathbf{1} R_f) \quad (6)'$$

From (7),

$$\begin{aligned}
\mu_p &= R_f + \tilde{w}_p^T (\mu - \mathbf{1}R_f) \\
&= R_f + \lambda \left[\Sigma^{-1} (\mu - \mathbf{1}R_f) \right]^T (\mu - \mathbf{1}R_f) \quad (\text{From (6)'}) \\
&= R_f + \lambda (\mu - \mathbf{1}R_f)^T \Sigma^{-1} (\mu - \mathbf{1}R_f)
\end{aligned}$$

$$\text{Let } H = (\mu - \mathbf{1}R_f)^T \Sigma^{-1} (\mu - \mathbf{1}R_f)$$

$$\text{then } \lambda = \frac{(\mu_p - R_f)}{H} \quad (8)$$

Plugging (8) into (6)' yields

$$\tilde{w}_p = \Sigma^{-1} (\mu - \mathbf{1}R_f) \frac{(\mu_p - R_f)}{H} \quad (9)$$

(optimal weight for n risky assets)

$$w_f = 1 - \tilde{w}_p^T \cdot \mathbf{1} \quad (\text{optimal weight for riskless asset})$$

*Variance of the frontier portfolio

$$\begin{aligned}
\sigma_p^2 &= \tilde{w}_p^T \Sigma \tilde{w}_p \\
\sigma_p^2 &= \frac{(\mu_p - R_f)}{H} (\mu - \mathbf{1}R_f)^T \Sigma^{-1} \Sigma \Sigma^{-1} (\mu - \mathbf{1}R_f) \frac{(\mu_p - R_f)}{H} \quad (\text{From (9)}) \\
&= \left(\frac{\mu_p - R_f}{H} \right)^2 \cdot H \\
&= \frac{(\mu_p - R_f)^2}{H}
\end{aligned}$$

$$\text{So } \sigma_p = \begin{cases} \frac{\mu_p - R_f}{\sqrt{H}} & \text{if } \mu_p > R_f \\ -\frac{\mu_p - R_f}{\sqrt{H}} & \text{if } \mu_p < R_f \end{cases}$$

- Note

$$\begin{aligned}
H &= (\mu - \mathbf{1}R_f)^T \Sigma^{-1} (\mu - \mathbf{1}R_f) \\
&= \mu^T \Sigma^{-1} \mu - 2R_f \cdot \mu^T \Sigma^{-1} \mathbf{1} + R_f^2 \cdot \mathbf{1}^T \Sigma^{-1} \mathbf{1} \\
&= B - 2R_f \cdot A + R_f^2 \cdot C
\end{aligned}$$

*** Investment weights for the tangency portfolio, m**

From (9), $\tilde{w}_m = \Sigma^{-1} (\mu - \mathbf{1}R_f) \left(\frac{\mu_m - R_f}{H} \right)$. But we do not know μ_m .

At the tangency portfolio, m , there is no riskless asset. That is,

$$w_f = 1 - \tilde{w}_m^T \cdot \mathbf{1} = 0 \Rightarrow \tilde{w}_m^T \cdot \mathbf{1} = 1$$

So $\tilde{w}_m^T \cdot \mathbf{1} = \left(\frac{\mu_m - R_f}{H} \right) (\mu^T - R_f \cdot \mathbf{1}^T) \Sigma^{-1} \cdot \mathbf{1}$ from (9)

$$\begin{aligned}
&= \left(\frac{\mu_m - R_f}{H} \right) (\mu^T \cdot \Sigma^{-1} \mathbf{1} - R_f \cdot \mathbf{1}^T \cdot \Sigma^{-1} \mathbf{1}) \\
&= \left(\frac{\mu_m - R_f}{H} \right) (A - R_f \cdot C) = 1
\end{aligned}$$

$$\Rightarrow \frac{\mu_m - R_f}{H} = \frac{1}{A - R_f \cdot C}$$

Now from (9),

$$\tilde{w}_m = \Sigma^{-1} (\mu - \mathbf{1}R_f) \left(\frac{\mu_m - R_f}{H} \right) = \frac{\Sigma^{-1} (\mu - \mathbf{1}R_f)}{A - R_f \cdot C} \quad (10)$$

(or) by normalization

$$\tilde{w}_m = \frac{\tilde{w}_p}{\tilde{w}_p^T \cdot \mathbf{1}}$$

$$\begin{aligned}\tilde{w}_m &= \frac{\tilde{w}_p}{\tilde{w}_p^T \cdot \mathbf{1}} = \frac{\Sigma^{-1}(\tilde{\mu} - \mathbf{1}R_f) \left(\frac{\mu_p - R_f}{H} \right)}{\left(\frac{\mu_p - R_f}{H} \right) (\tilde{\mu} - \mathbf{1}R_f)^T \Sigma^{-1} \cdot \mathbf{1}} = \frac{\Sigma^{-1}(\tilde{\mu} - \mathbf{1}R_f)}{(\tilde{\mu} - \mathbf{1}R_f)^T \Sigma^{-1} \cdot \mathbf{1}} \\ &= \frac{\Sigma^{-1}(\tilde{\mu} - \mathbf{1}R_f)}{\tilde{\mu}^T \cdot \Sigma^{-1} \mathbf{1} - R_f \cdot \mathbf{1}^T \cdot \Sigma^{-1} \mathbf{1}} = \frac{\Sigma^{-1}(\tilde{\mu} - \mathbf{1}R_f)}{A - R_f \cdot C}\end{aligned}$$

*An example of normalization

e.g. Two risky asset plus one riskless asset

Suppose $\tilde{w}_p = \begin{pmatrix} 0.40 \\ 0.50 \end{pmatrix}$, $w_f = 0.1$. What is $\tilde{w}_m$?

By normalization two weights for the optimal risky portfolio, m ,

$$\begin{aligned}w_1^* &= \frac{w_1}{\sum w_i} = \frac{0.40}{0.4 + 0.5} = \frac{4}{9} \\ w_2^* &= \frac{w_2}{\sum w_i} = \frac{0.50}{0.4 + 0.5} = \frac{5}{9}\end{aligned}$$

*** Mean and Variance of the tangency portfolio, m**

$$\begin{aligned}\mu_m &= \tilde{\mu}^T \tilde{w}_m \\ &= \tilde{\mu}^T \cdot \frac{\Sigma^{-1}(\tilde{\mu} - \mathbf{1}R_f)}{A - R_f \cdot C} = \frac{\tilde{\mu}^T \Sigma^{-1} \tilde{\mu} - R_f \cdot \tilde{\mu}^T \Sigma^{-1} \mathbf{1}}{A - R_f \cdot C} \quad ((\text{From (10)}) \\ &= \frac{B - R_f \cdot A}{A - R_f \cdot C}\end{aligned} \tag{11}$$

$$\begin{aligned}\sigma_m^2 &= \tilde{w}_m^T \Sigma \tilde{w}_m \\ &= \frac{(\tilde{\mu} - \mathbf{1}R_f)^T \Sigma^{-1} \Sigma \Sigma^{-1} (\tilde{\mu} - \mathbf{1}R_f)}{(A - R_f \cdot C)^2}\end{aligned}$$

$$= \frac{H}{(A - R_f \cdot C)^2} \quad (12)$$

***Special case that $R_f = \frac{A}{C}$**

From (11) and (12), when $R_f = \frac{A}{C}$, $\mu_m \Rightarrow \pm\infty$ and $\sigma_m \Rightarrow \pm\infty$, which means that the tangency portfolio close not exist.

The slope of the line is

$$\begin{aligned} \sqrt{H} &= (B - 2R_f \cdot A + R_f^2 \cdot C)^{\frac{1}{2}} \\ &= \left(B - 2\left(\frac{A}{C}\right) \cdot A + \left(\frac{A}{C}\right)^2 \cdot C \right)^{\frac{1}{2}} \\ &= \left(\frac{BC - A^2}{C} \right)^{\frac{1}{2}} = \left(\frac{D}{C} \right)^{\frac{1}{2}} \end{aligned}$$

Thus, the line is $\mu = \frac{A}{C} \pm \sqrt{\frac{D}{C}} \cdot \sigma$

These are the asymptotes of the portfolio frontier with risky assets only.

In this case, the sum of the weights for the tangency portfolio is

$$\begin{aligned} \tilde{w}_p^T \cdot \mathbf{1} &= \left(\frac{\mu_p - R_f}{H} \right) (\tilde{\mu}^T - R_f \cdot \mathbf{1}^T) \Sigma^{-1} \cdot \mathbf{1} \\ &= \left(\frac{\mu_p - R_f}{H} \right) (\tilde{\mu}^T \Sigma^{-1} \mathbf{1} - R_f \mathbf{1}^T \Sigma^{-1} \mathbf{1}) = \left(\frac{\mu_p - R_f}{H} \right) (A - R_f \cdot C) \\ &= \left(\frac{\mu_p - R_f}{H} \right) \left(A - \left(\frac{A}{C} \right) \cdot C \right) = 0 \quad (\text{arbitrage portfolio}) \end{aligned}$$

indicating that an investor put 0% in the risky assets and 100% in riskless asset.

***Covariance between a frontier portfolio, p , and any portfolio, q**

$$\begin{aligned}
 \sigma_{pq} &= \tilde{w}_q^T \Sigma \tilde{w}_p \\
 &= \tilde{w}_q^T \Sigma \left[\Sigma^{-1} \left(\tilde{\mu} - \mathbf{1} R_f \right) \left(\frac{\mu_p - R_f}{H} \right) \right] && (\text{note } \tilde{w}_q^T \cdot \mathbf{1} \neq 1) \\
 &= \left(\tilde{w}_q^T \tilde{\mu} - R_f \cdot \tilde{w}_q^T \cdot \mathbf{1} \right) \left(\frac{\mu_p - R_f}{H} \right) \\
 &= \frac{(\mu_q - R_f)(\mu_p - R_f)}{H} && (13)
 \end{aligned}$$

Note

$$\sigma_p^2 = \frac{(\mu_p - R_f)^2}{H}$$

Remark

$$\begin{aligned}
 \because \mu_q &= \tilde{w}_q^T \tilde{\mu} + (1 - \tilde{w}_q^T \mathbf{1}) \cdot R_f \\
 &= \tilde{w}_q^T \tilde{\mu} + R_f - R_f \tilde{w}_q^T \mathbf{1} \\
 \Rightarrow \mu_q - R_f &= \tilde{w}_q^T \tilde{\mu} - R_f \tilde{w}_q^T \mathbf{1}
 \end{aligned}$$

* Derivation of the CAPM (Capital Asset Pricing Model)

Let q be any portfolio (not necessarily frontier portfolio) or any individual asset.

Let P be a frontier portfolio with the portfolio weight, $\tilde{w}_p$, on risky assets ($\mu_p \neq R_f$)

$$\begin{aligned}\sigma_{pq} &= \tilde{w}_q' \Sigma \tilde{w}_p \\ &= \frac{(\mu_q - R_f)(\mu_p - R_f)}{H} \quad (\text{From (13)})\end{aligned}$$

$$\begin{aligned}\Rightarrow \mu_q - R_f &= \frac{\sigma_{pq}}{\frac{(\mu_p - R_f)}{H}} = \frac{\sigma_{pq}}{\frac{(\mu_p - R_f)^2}{H}} (\mu_p - R_f) \\ &= \frac{\sigma_{pq}}{\sigma_p^2} (\mu_p - R_f)\end{aligned}$$

If $p=m$ (market portfolio), then

$$\mu_q = R_f + \beta_{qm} (\mu_m - R_f) \quad \text{where} \quad \beta_{qm} = \frac{\sigma_{qm}}{\sigma_m^2}$$